



**Interoperability among different systems in smart city
using blockchain.**



Contents

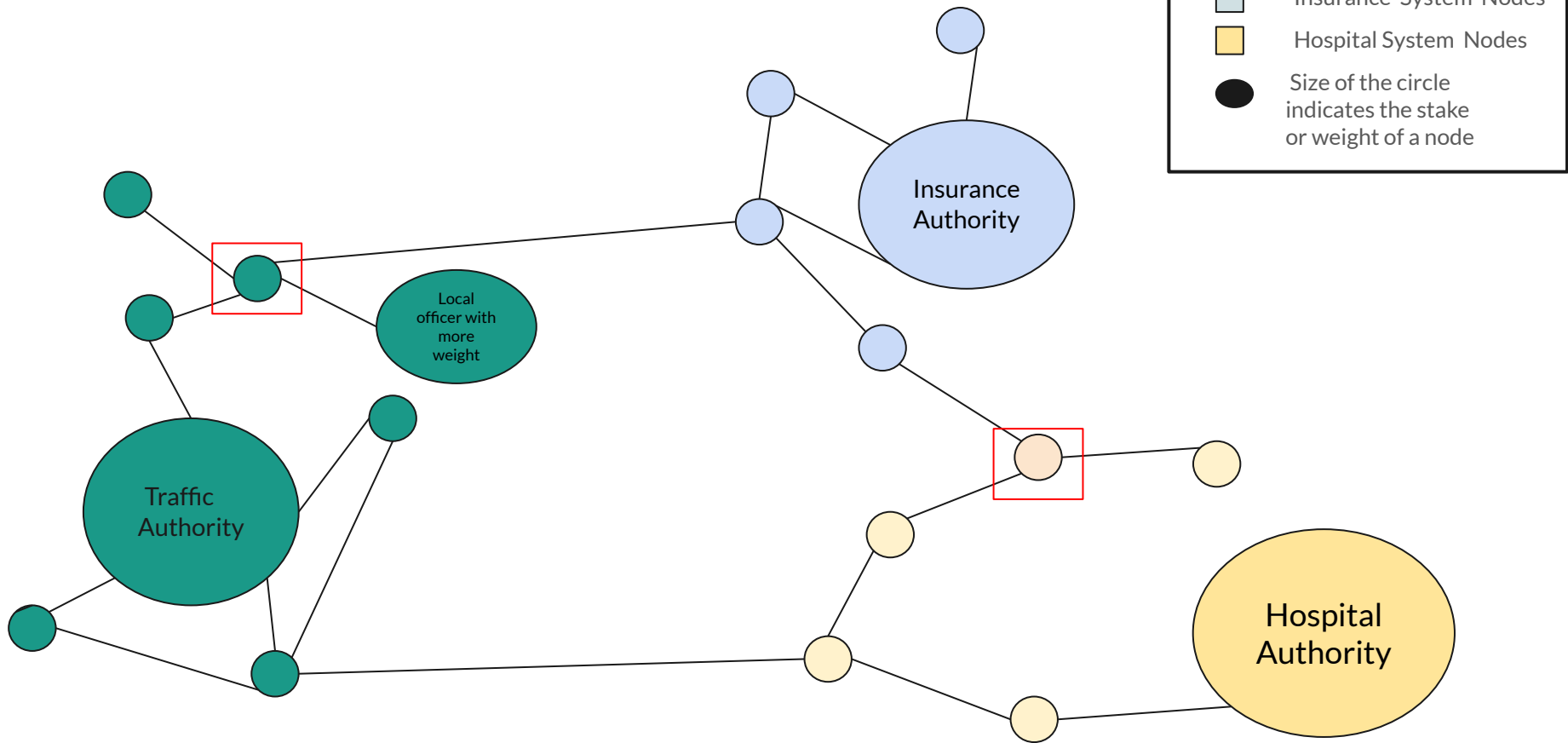
1. The Hybrid Consensus Mechanism - Neighbour Before Authority
 - a) Communications in NBA model
 - b) Working of NBA Model
2. Distributed Ledger Technology - IOTA
 - a) Tip Selection Algorithms
 - b) Arising problems
 - c) Implementation
3. Future Scope
4. References



Neighbour Before Authority (NBA)

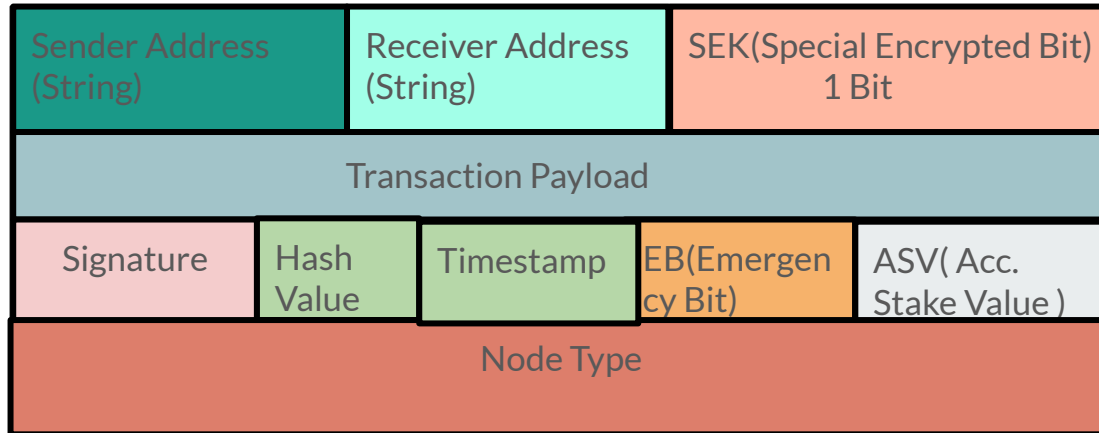
THE HYBRID CONSENSUS MECHANISM

An Example of the Hybrid Approach Verification





Transaction Structure





Communications in the NBA Model

- Communications in the NBA model happen in the form of messages between peers like in any decentralised Network.
- The communications happening in the NBA Model are
 - OFF Ledger Communications
 - ON Ledger Communications



OFF Ledger Communications

- The communications which happen in the network **between the node of the same type of system**
- They happen between the local nodes and between the initiator node and authority of the system in the times of need.
- The main aim of this type of communication happens to accumulate the stake needed to get the transaction verified.
- Once Verified the OFF Ledger communications ends laying ground for the ON Ledger Communication to get into the Picture.



Two cases Arise in the OFF Ledger Transaction

i)After Appending the stake the Neighbouring Nodes send the transaction to the src node ,

Now the source node checks the updates stake value and decides to whether send it to the authority or to append it to the ledger and broadcast the successful addition to the Ledger.

ii)Parallellly while initiating the transaction the timestamp is recorded and is an integral part of the transaction as it proceeds in the network, if the source doesn't get enough stake in the given time or the verifying neighbour notices that the timestamp upon exceeding the threshold the transaction will be redirected to the authority.



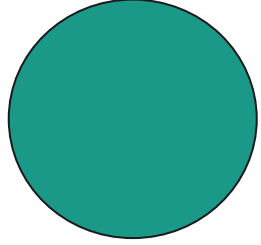
ON Ledger Communication

- ON Ledger communication is the term coined for the messages that become part of the ledger upon meeting the verification policies.
- These communication are the **typical Transactions** of the Traditional blockchain that are to be pushed into the ledger maintained by each node.
- The off ledger message are transformed into the ON ledger Transaction , initiated by other genesis node of the transaction or the authority according to the respective case.
- Each peer will be broadcasting the Transaction to its peers, after appending the transaction to their ledgers.

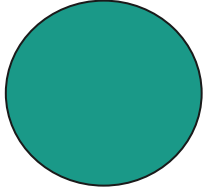


THE TRAFFIC INCIDENT

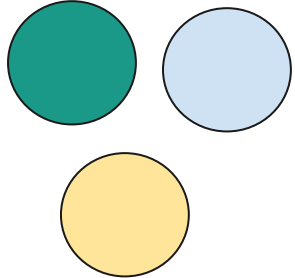
An Example Scenario to look into the deeper levels of the working of the NBA Algo.



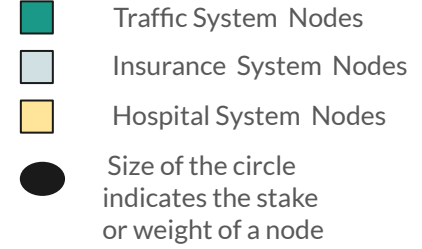
High level Delegate .Node with adequate Computational power and in the pinnacle of the hierarchy of the system organisation.



Middle level Delegates in each System .
For eg :-District Traffic corporation Head.

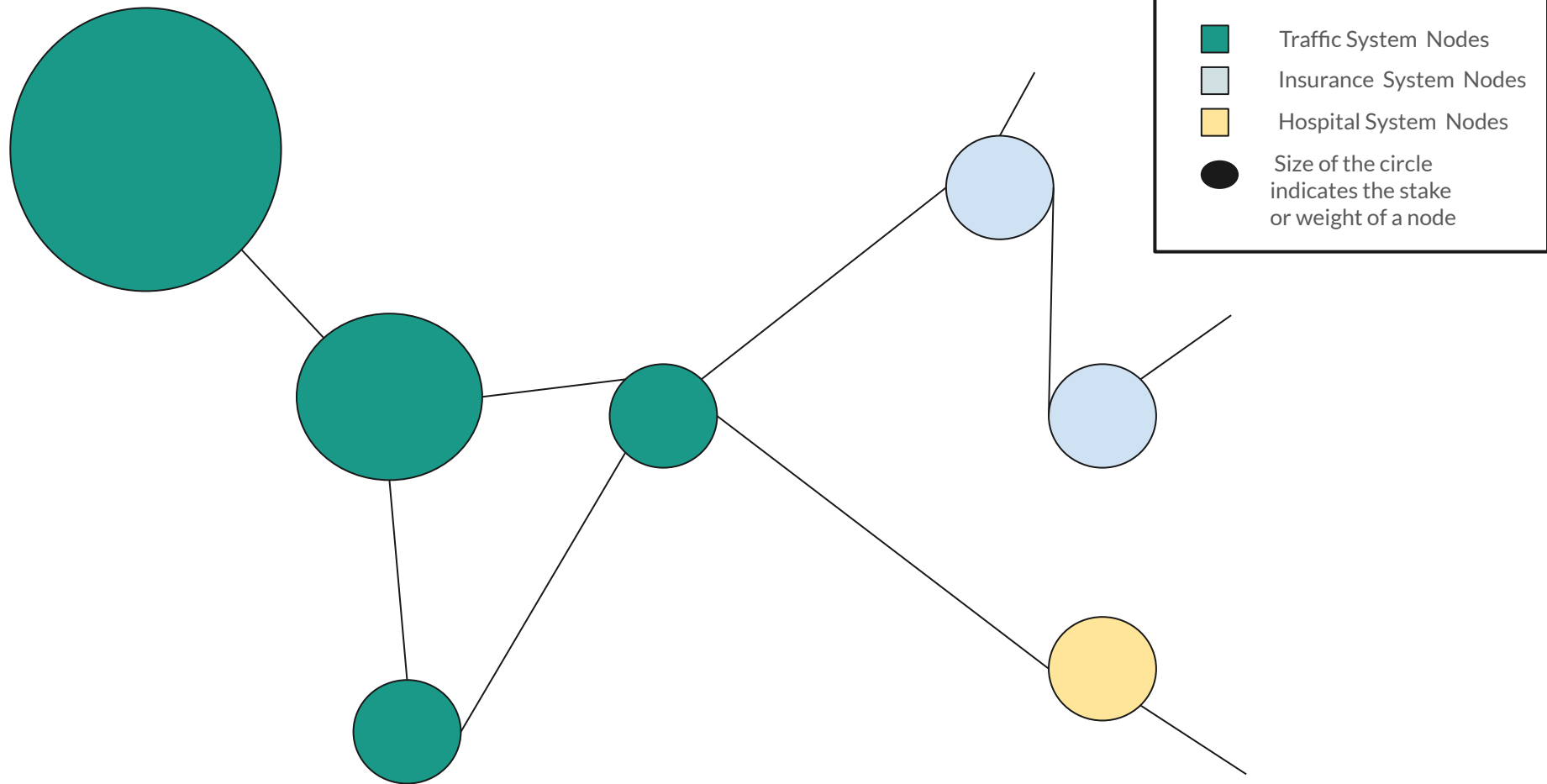


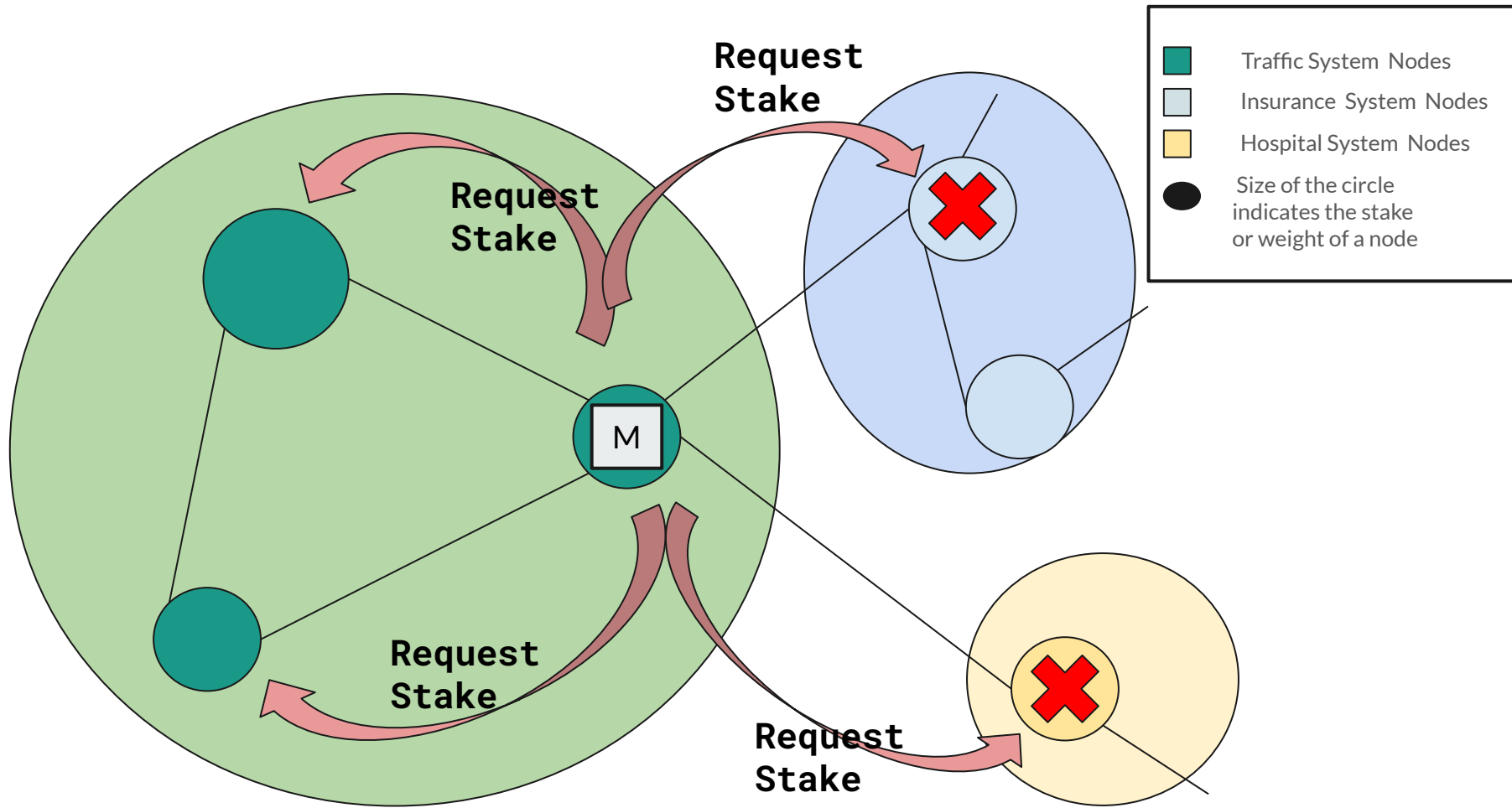
Low Level Delegates in each system,
For eg : - Traffic Police ,Insurance Collector,Ward Boy.

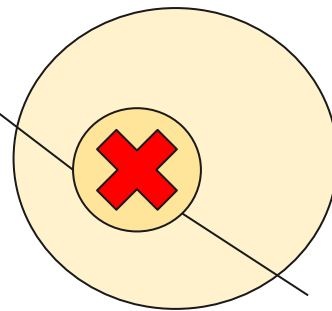
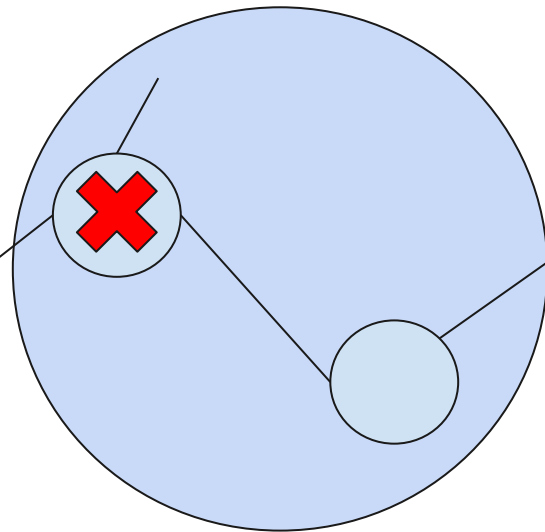
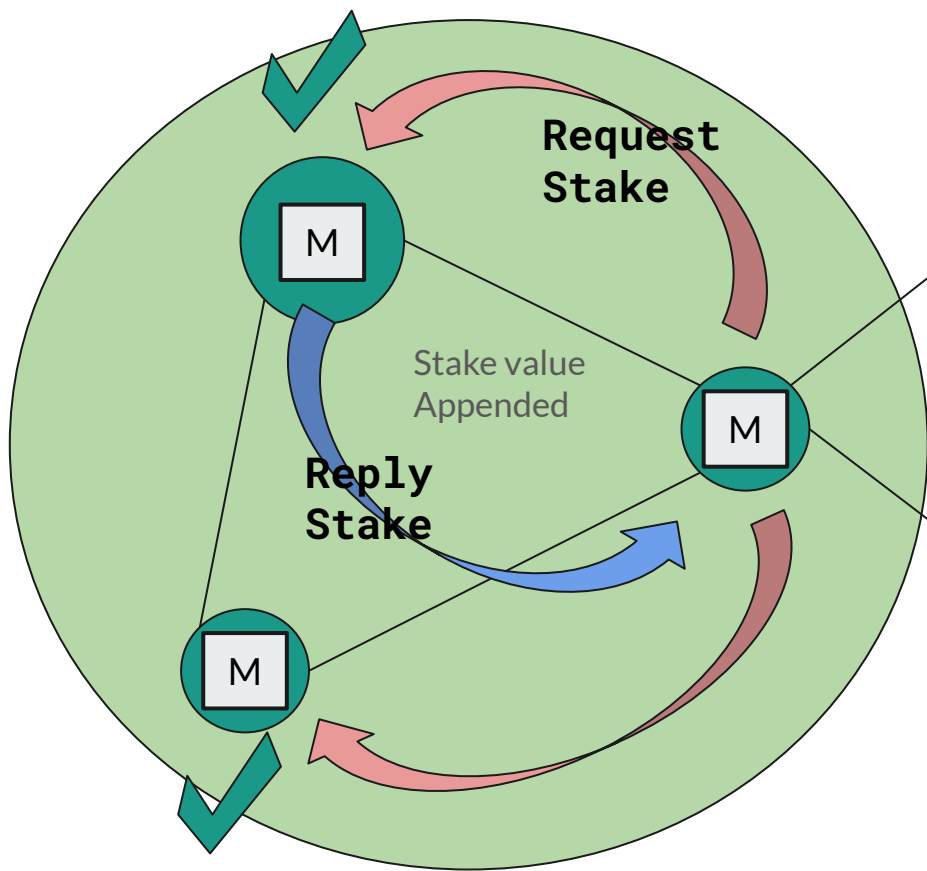


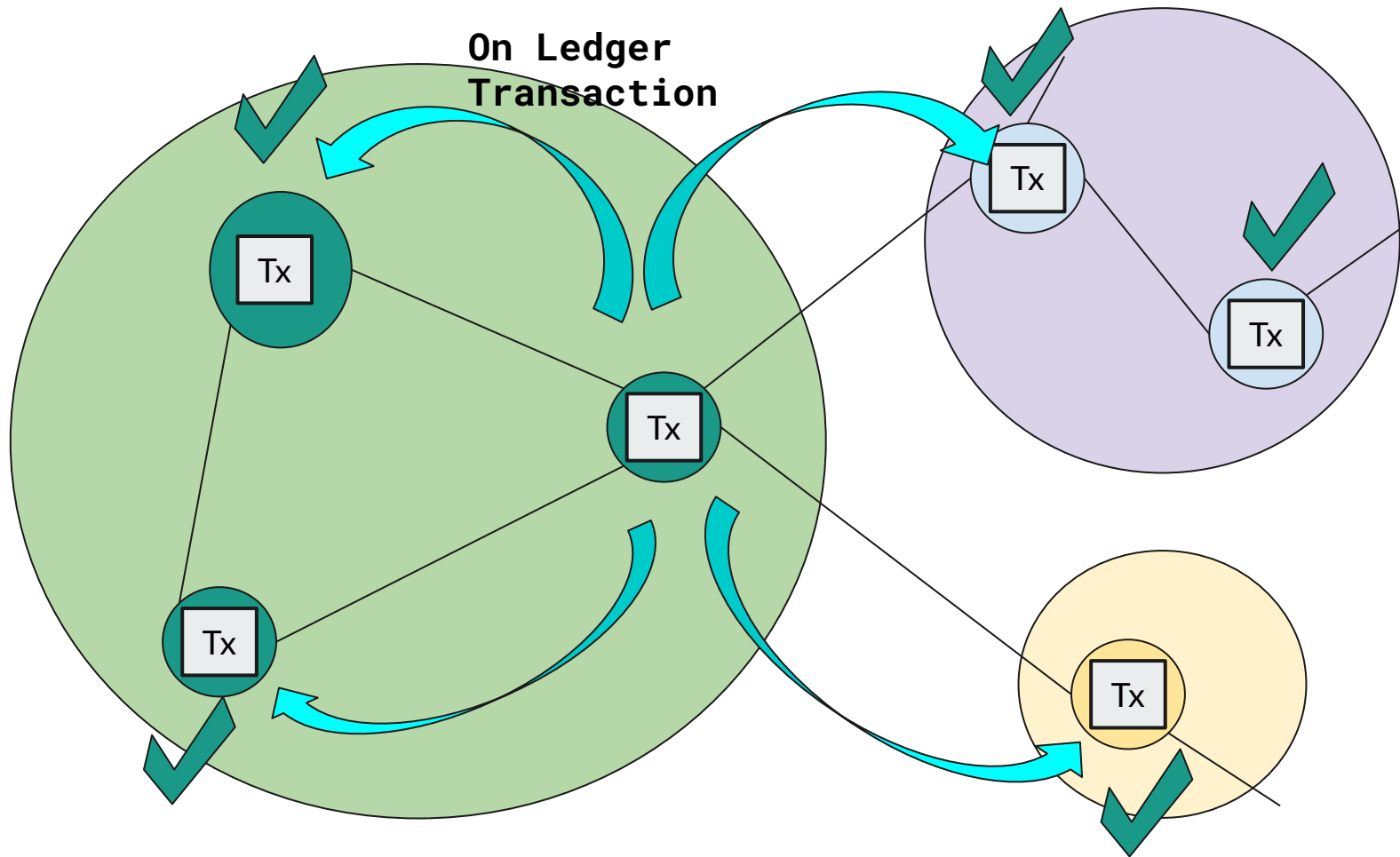


Case - I



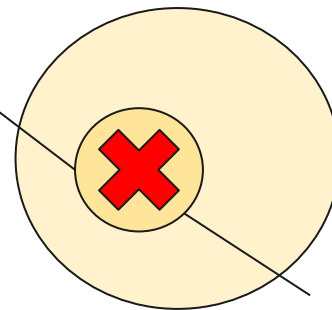
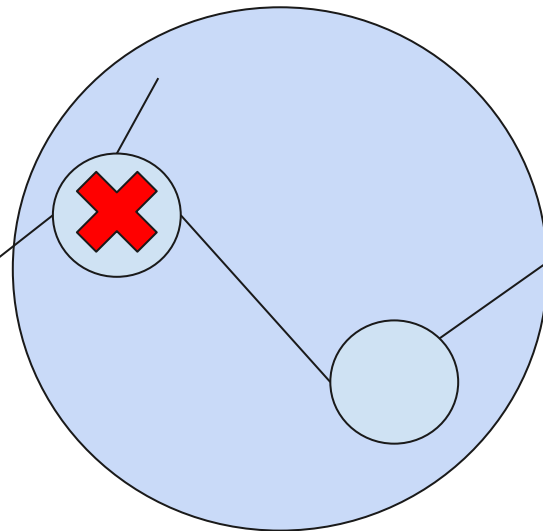
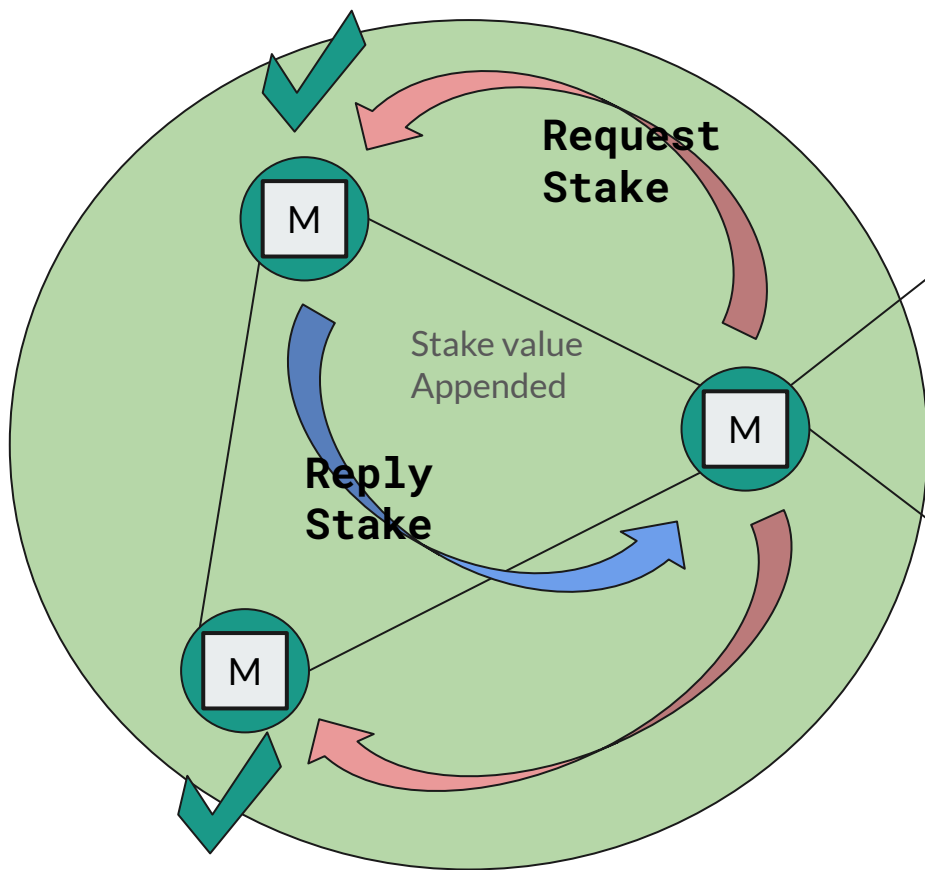


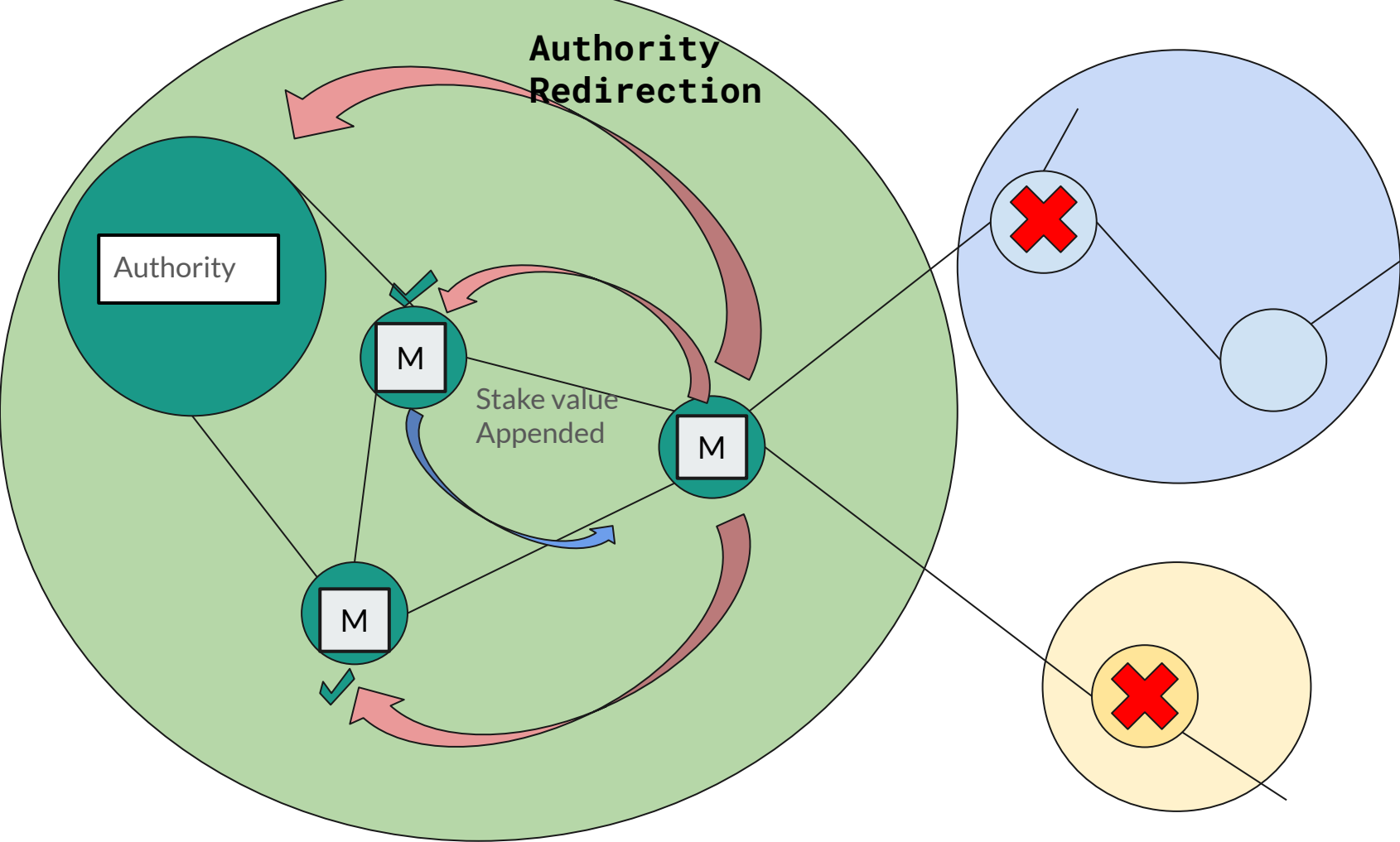


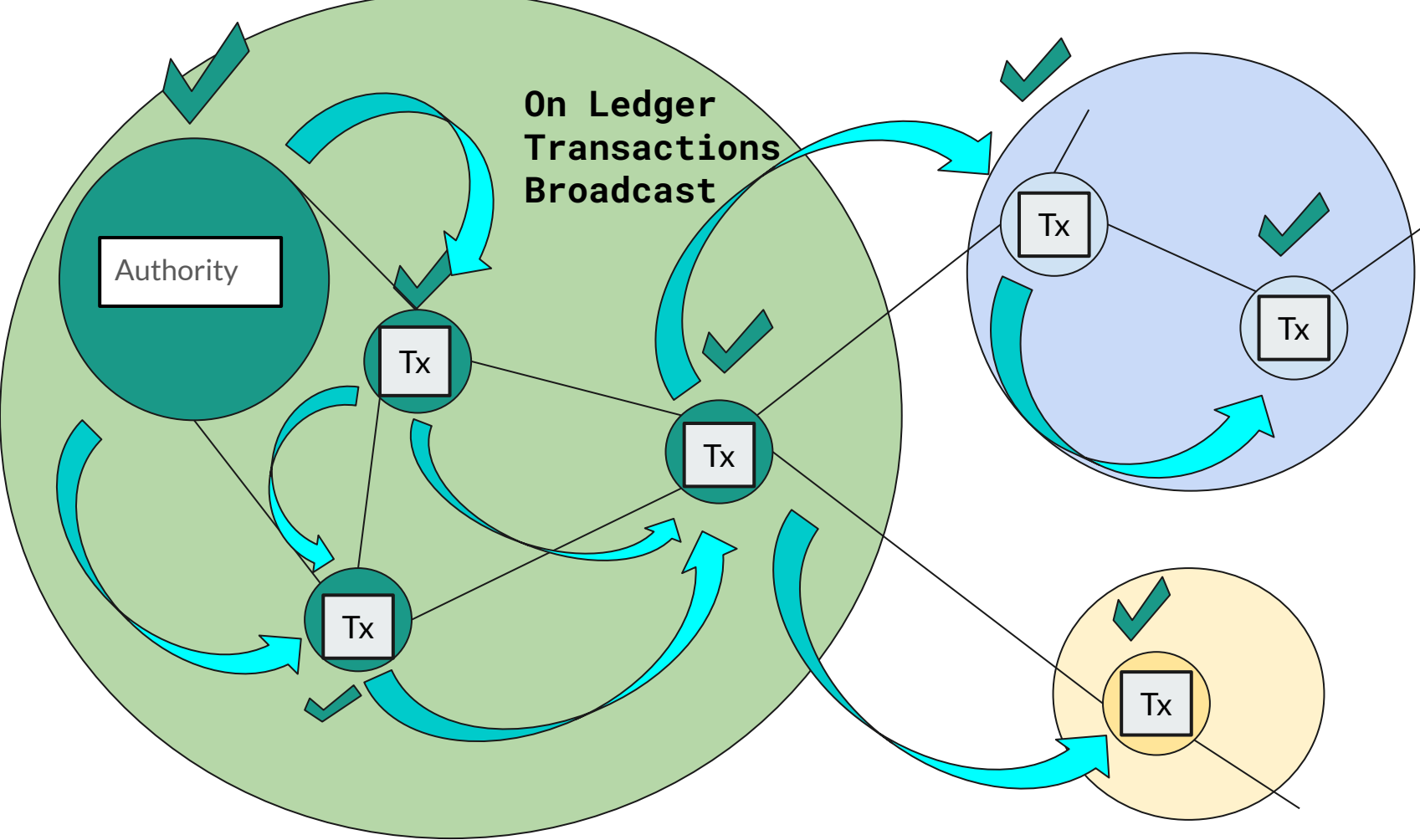




Case - II









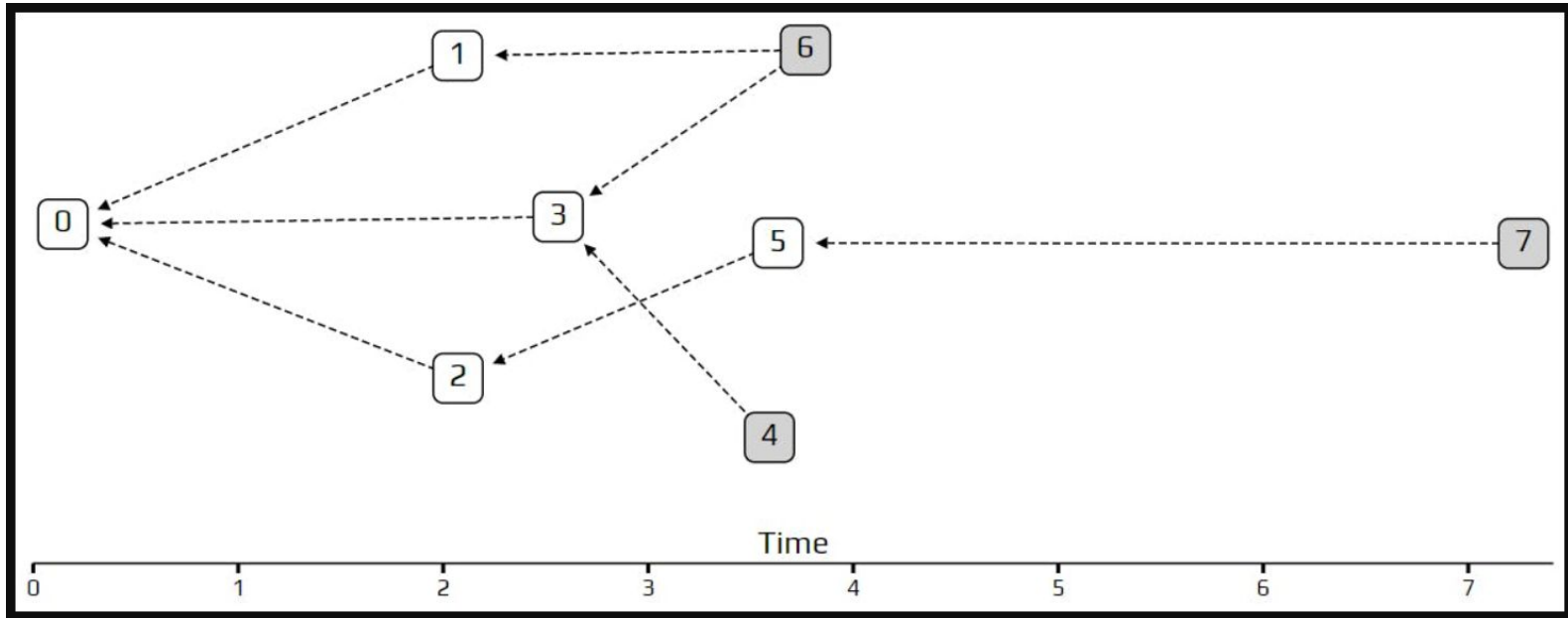
DAG



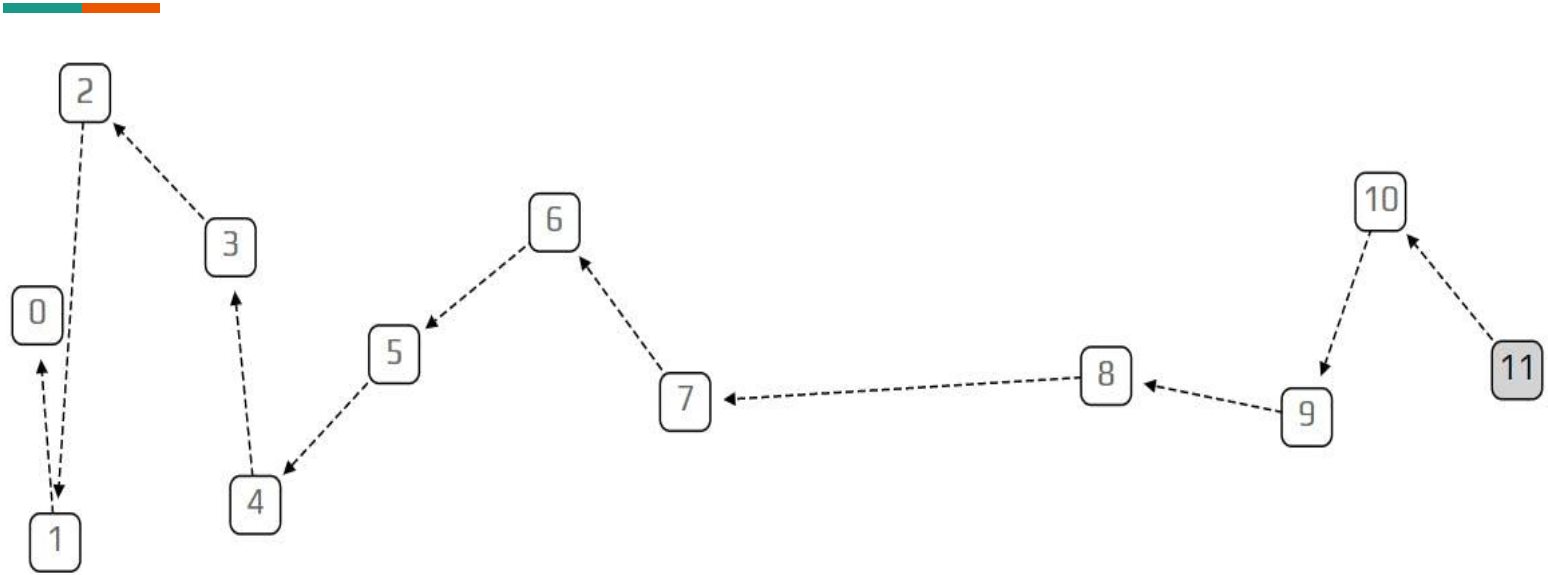
IOTA

- The Tangle, which is the data structure behind IOTA, is a particular kind of directed graph, which holds *transactions*
- Each transaction is represented as a vertex in the graph. When a new transaction joins the tangle, it chooses two previous transactions to *approve*
- Unapproved transactions *tips*
- The strategy for choosing which two tips to approve is very important, and is the key to IOTA's unique technology

Transaction Rates and Network Latency



- Poisson point process models how transactions should arrive

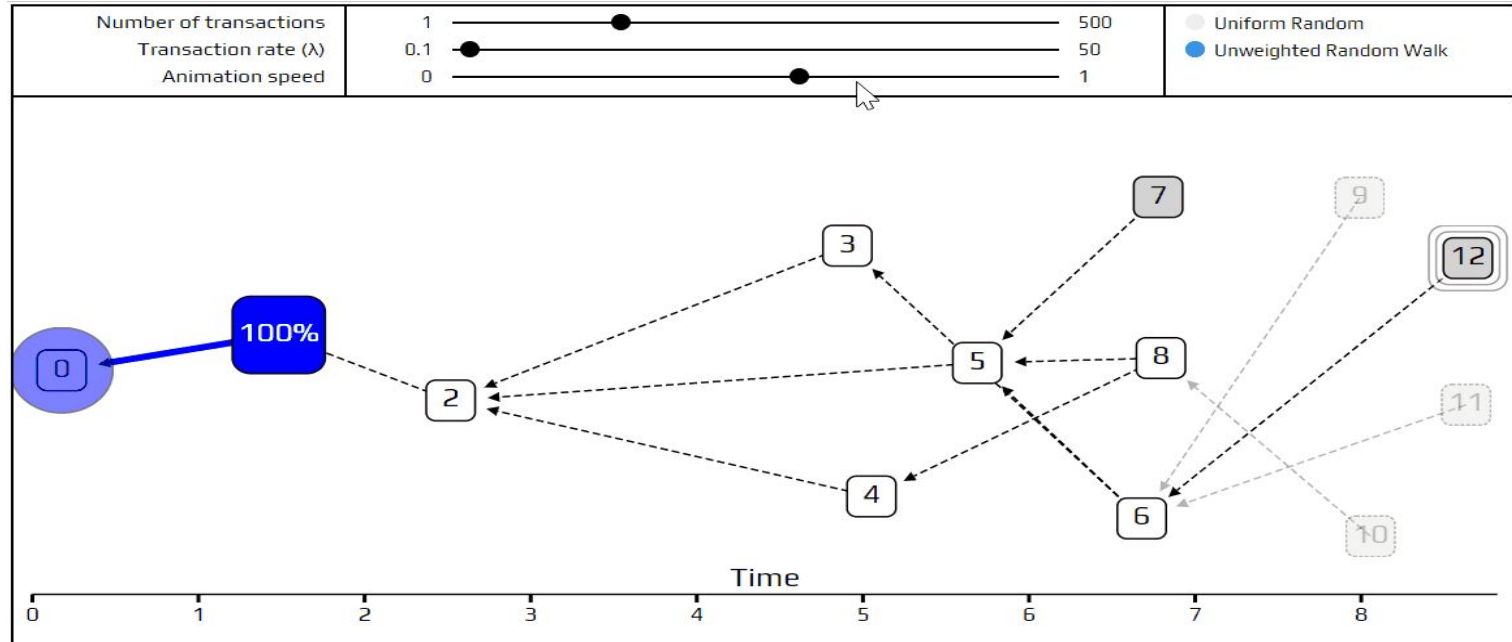


DAG behaves like traditional blockchain when the rate of incoming transactions is very small



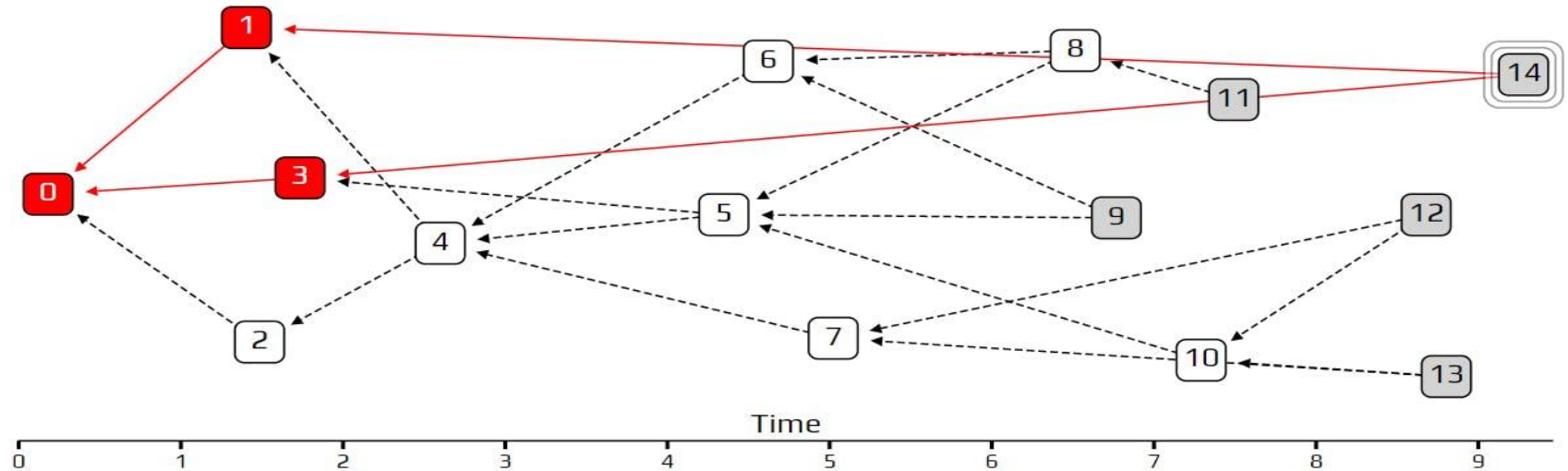
Tip Selection Algorithms

Random Walk

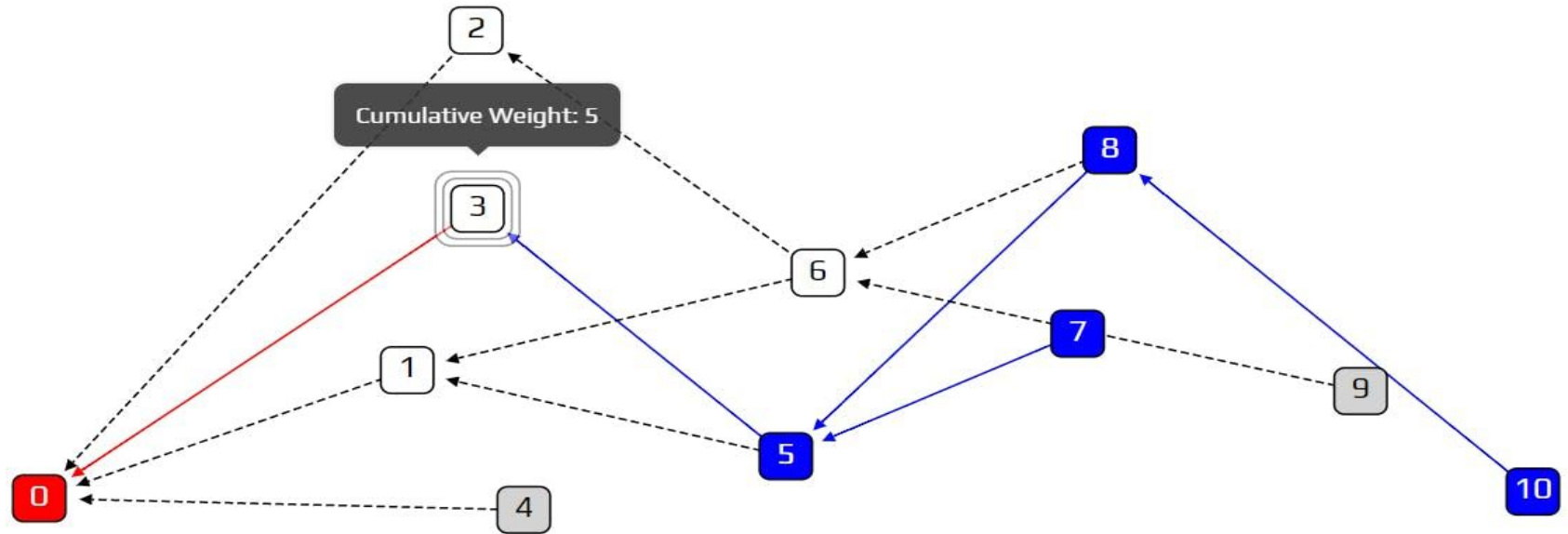


At each step it goes to one of the transactions which directly approved the current transaction and chooses the transaction with equal probability

Weighted Random Walk and Cumulative Weight



A lazy tip approves old transactions rather than recent ones. It will only broadcasts its own transactions based on old data.



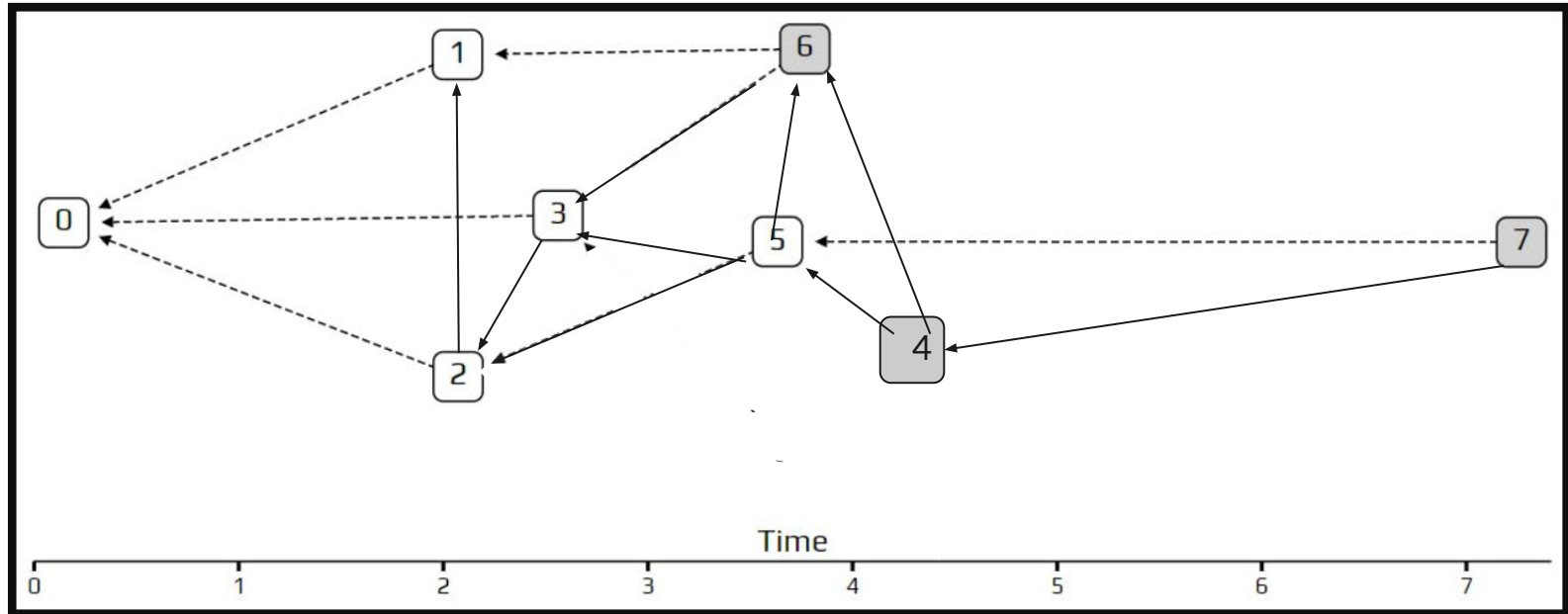
- Strategy will be to *bias* our random walk, so we are less likely to choose lazy tips.
- Cumulative weight is count how many approvers a transaction has, and add one.



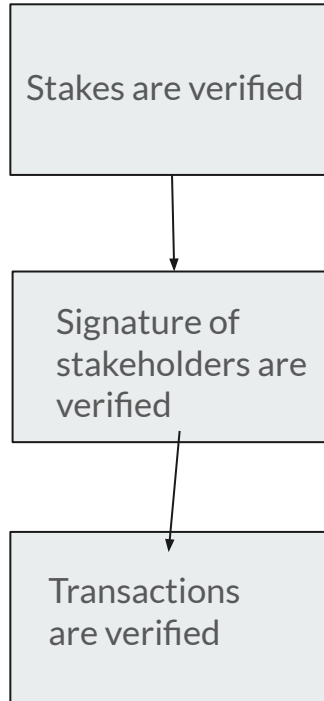
Here the problem arises

How the recent transactions(tip) will get verified?
How the DAG structure would be maintained?

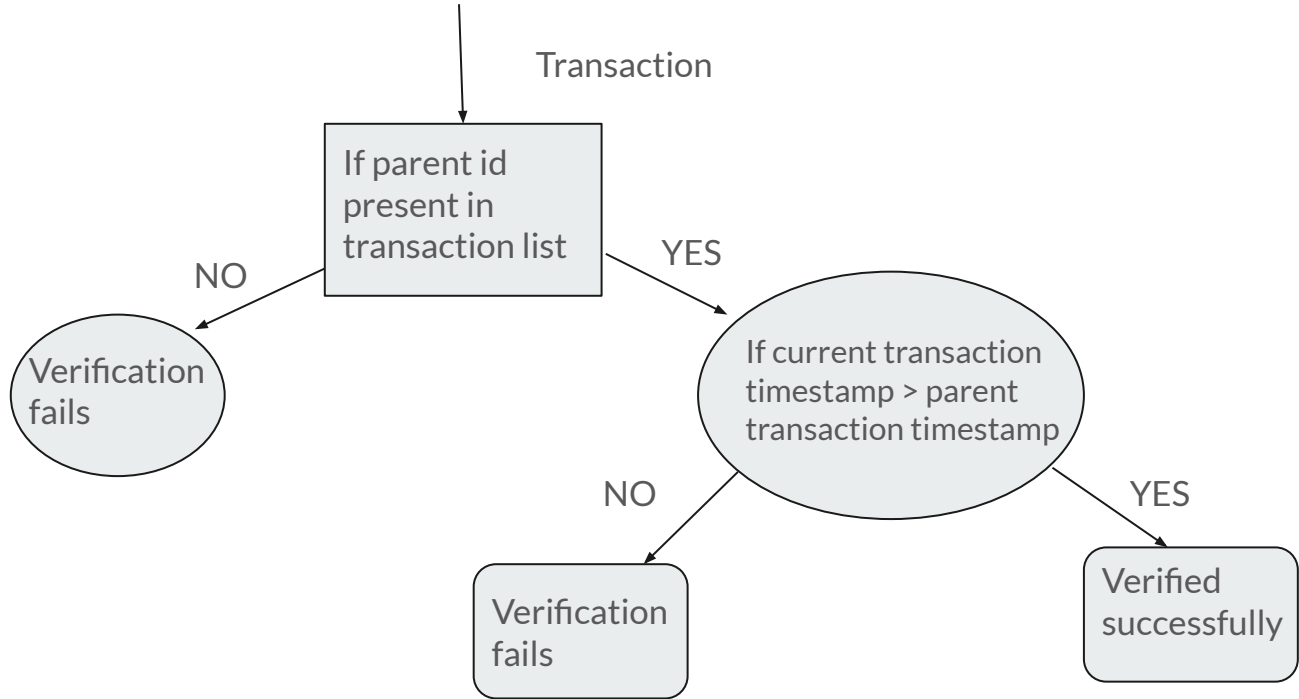
- 1.** Add subsequent transactions with random 2 parents from the latest 5 (Based on the Timestamp)



2. A) Verify the last transaction



2. b) Verify the DAG Structure

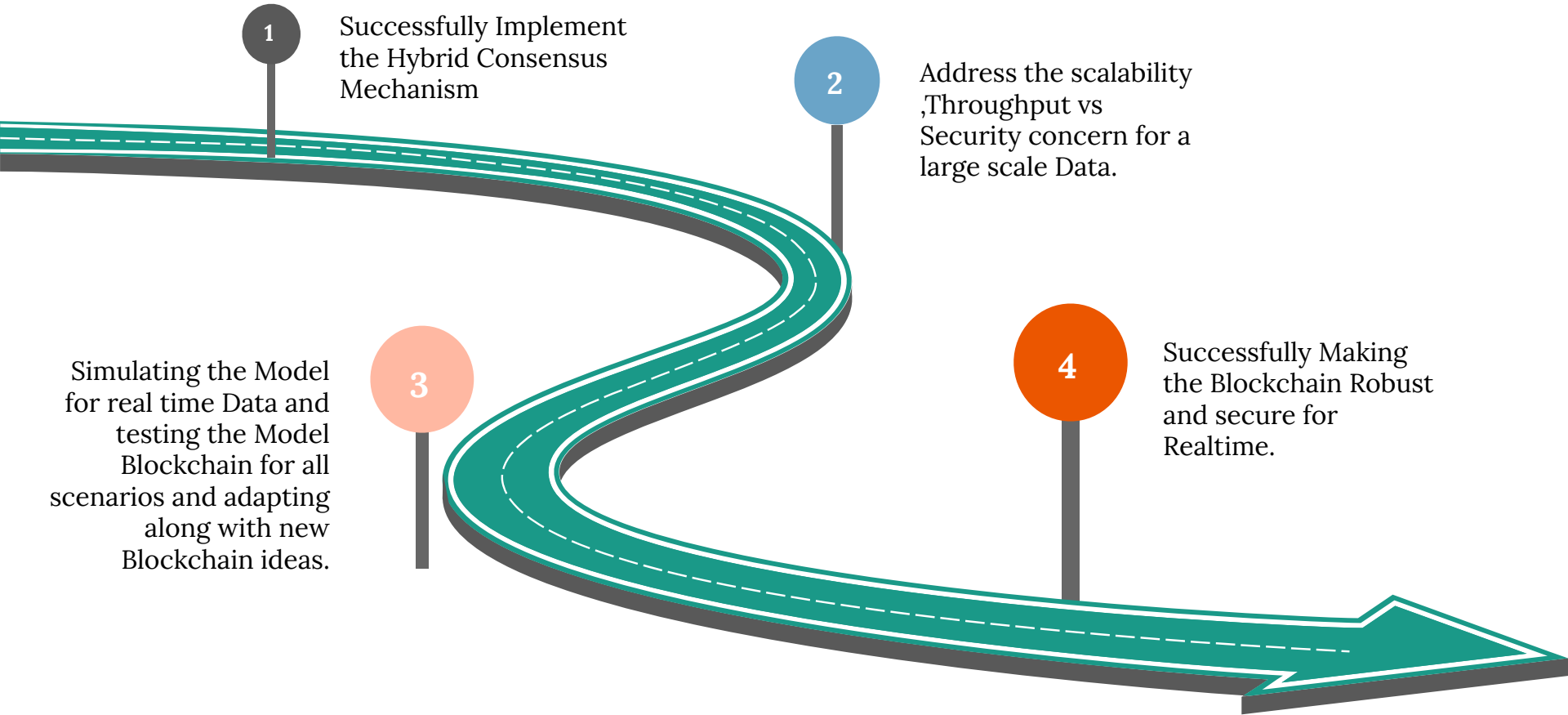




Implementation till now

1. Implementing the NBA consensus algo in the peer to peer till the local level and sequential stake collection is implemented .
2. The initiative to implement the parallel stake collection has been taken and implemented in a pseudo manner with the timestamp.
3. Network verification has been partial implemented.
4. Implemented the DAG structure using modified tip selection algorithm.
5. DAG Transaction verification has been partially implemented.

Roadmap To the Future





References

1. Pervez, H., Muneeb, M., Irfan, M.U. and Haq, I.U., 2018, December. A comparative analysis of DAG-based blockchain architectures. In *2018 12th International conference on open source systems and technologies (ICOSST)* (pp. 27-34). IEEE.
 - Blockchain Technology Concepts and Applications by Kumar Saurabh and Ashutosh Saxena
 - Raj J.M., Ranjani S.S.
A secured blockchain method for multivariate industrial IoT-oriented infrastructure based on deep residual squeeze and excitation network with single candidate optimizer
Int. Things, 22 (2023), Article 100823
 - Zhao, Zhenwei, et al. "Secure internet of things (IoT) using a novel brooks lyengar quantum byzantine agreement-centered blockchain networking (BIQBA-BCN) model in smart healthcare." *Information Sciences* 629 (2023): 440-455.
 - ailshery, L.S. IoT Connected Devices Worldwide 2019–2030|Statista. 2022. Available online: <https://www.statista.com/statistics/1183457/iot-connected-devices-worldwide/>.
 - SUI official Documentation <https://docs.sui.io/>



Thank You

No of Times we met Dr Prashant Sir since the last sem fyp eval to this eval = **6**